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PRN- 23070122155

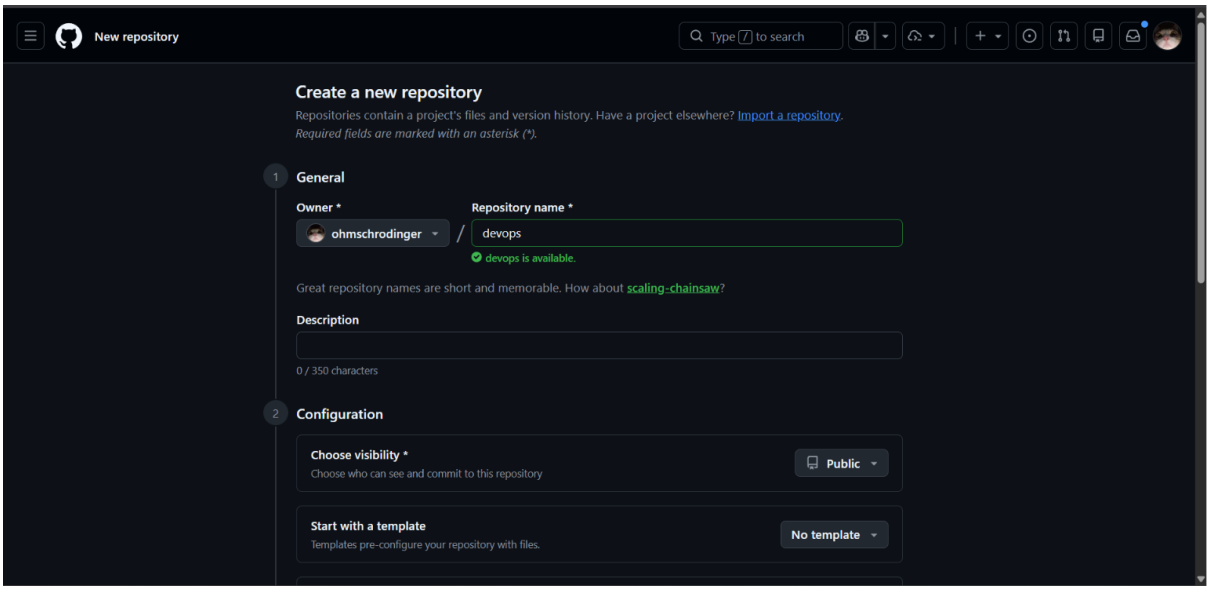
DevOps Lab Assessment Questions

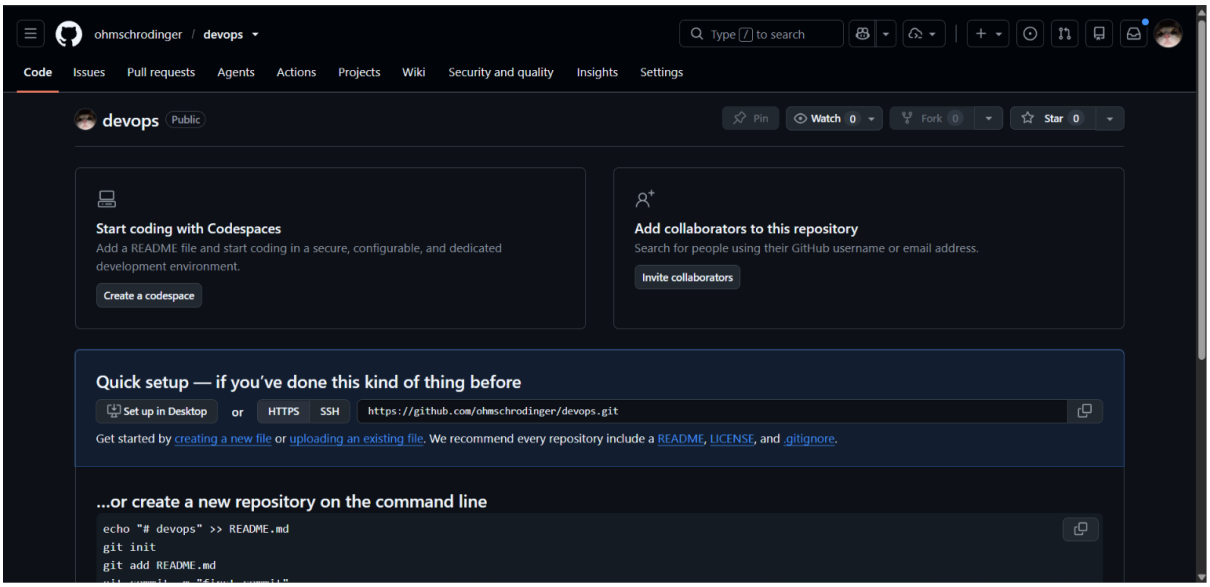
1. TW1 - Continuous Evaluation: Foundational DevOps Skills (10 Marks)

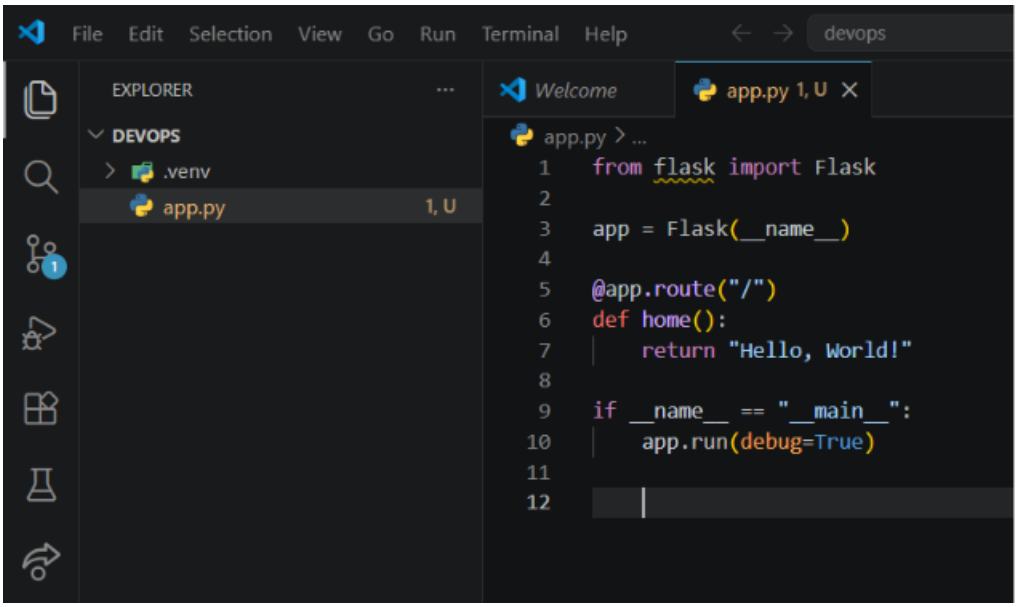
Instructions: Complete the following tasks in your lab environment. Document your commands and outputs as evidence.

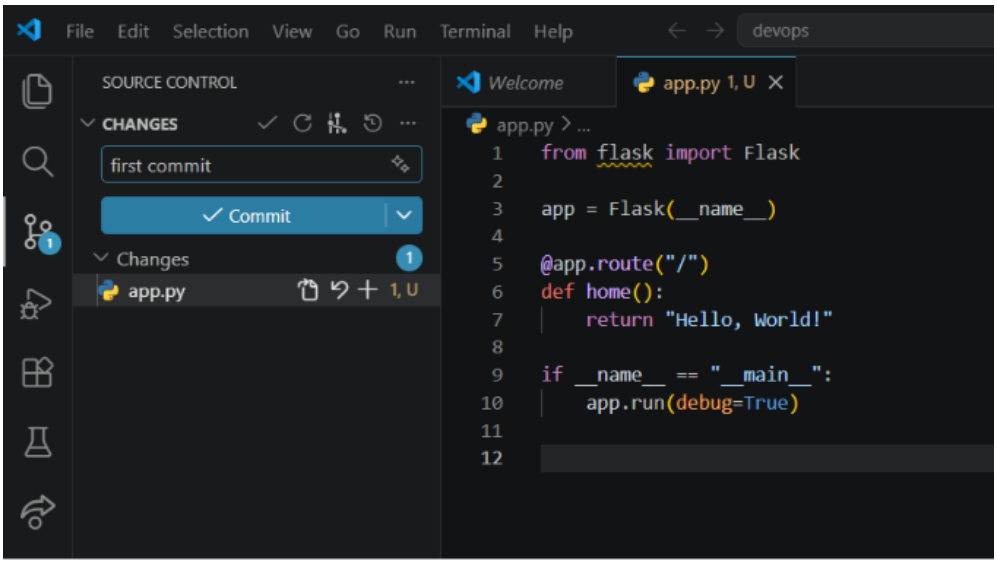
Scenario: You are a developer working on a new feature for a web application.

1. **Git Workflow & Collaboration (4 Marks)**
    - **Task 1.1:** Initialize a new Git repository for a simple "Hello World" Python Flask application (provided to you). Add the initial code and commit it to the main branch. (1 Mark)

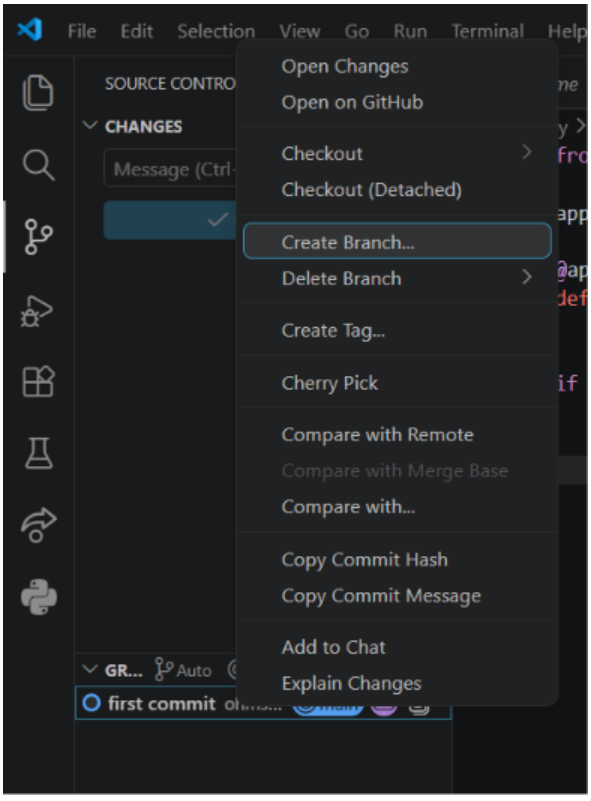


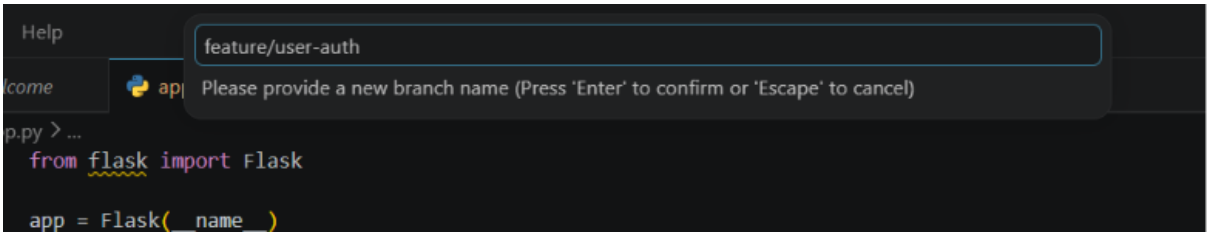

ohmschrodinger / devops
Type / to search
Code
Issues
Pull requests
Agents
Actions
Projects
Wiki
Security and quality
Insights
Settings
devops
Public
Pin
Watch 0
Fork 0
Star 0
Start coding with Codespaces
Add a README file and start coding in a secure, configurable, and dedicated development environment.
Create a codespace
Add collaborators to this repository
Search for people using their GitHub username or email address.
Invite collaborators
Quick setup — if you've done this kind of thing before
Set up in Desktop
or
HTTPS
SSH
https://github.com/ohmschrodinger/devops.git
Get started by creating a new file or uploading an existing file. We recommend every repository include a README, LICENSE, and .gitignore.
...or create a new repository on the command line
echo "# devops" >> README.md
git init
git add README.md

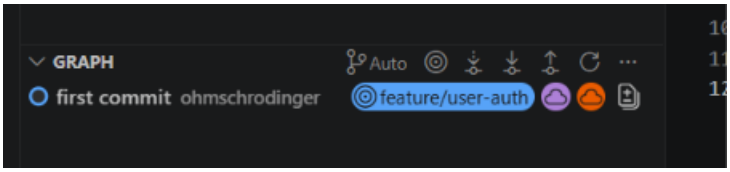

File
Edit
Selection
View
Go
Run
Terminal
Help
devops
EXPLORER
DEVOPS
.venv
app.py
1, U
Welcome
app.py 1, U
app.py > ...
1 from flask import Flask
2
3 app = Flask(__name__)
4
5 @app.route("/")
6 def home():
7 return "Hello, World!"
8
9 if __name__ == "__main__":
10 app.run(debug=True)
11
12

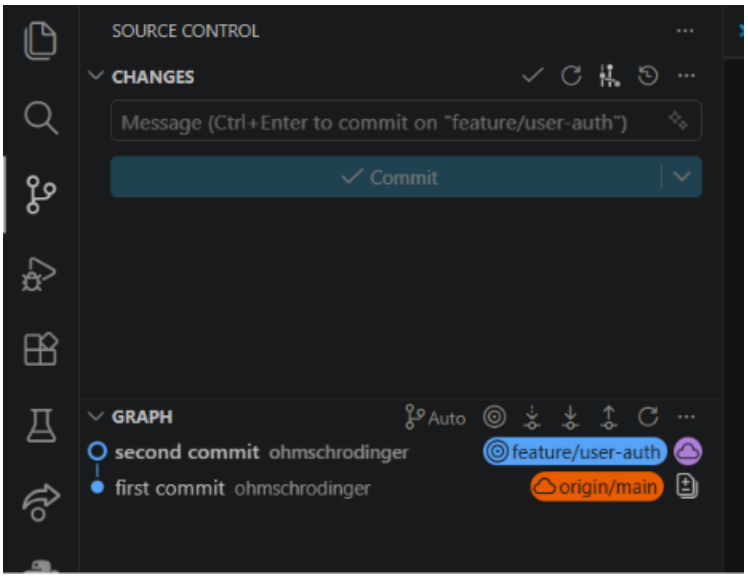


- **Task 1.2:** Create a new branch named feature/user-auth. Make a small modification (e.g., add a new print statement) in this branch, commit the changes, and push the branch to a remote repository (e.g., GitHub/GitLab). (1.5 Marks)

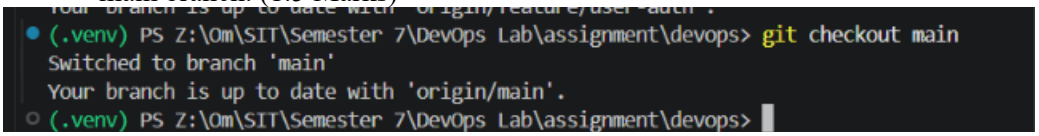


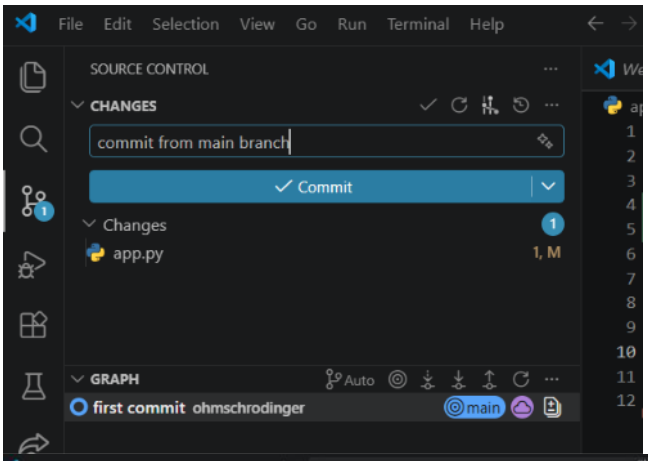


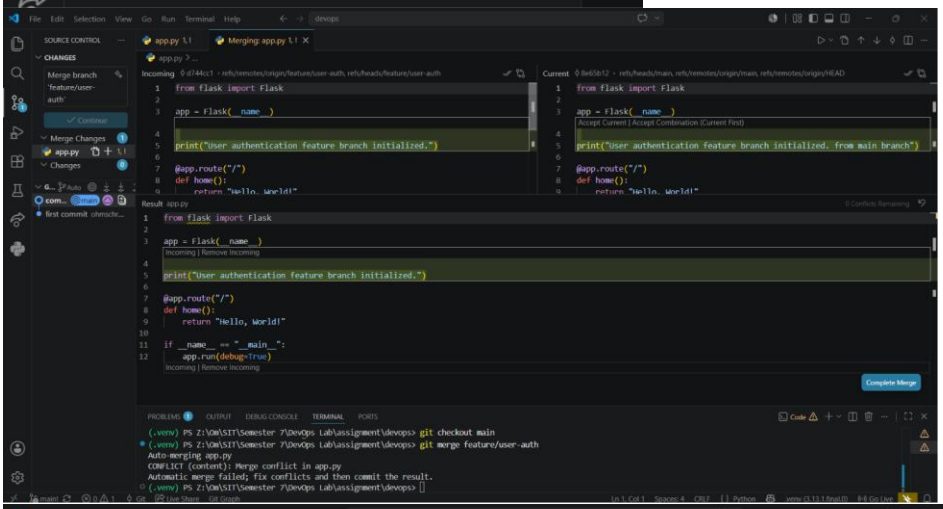


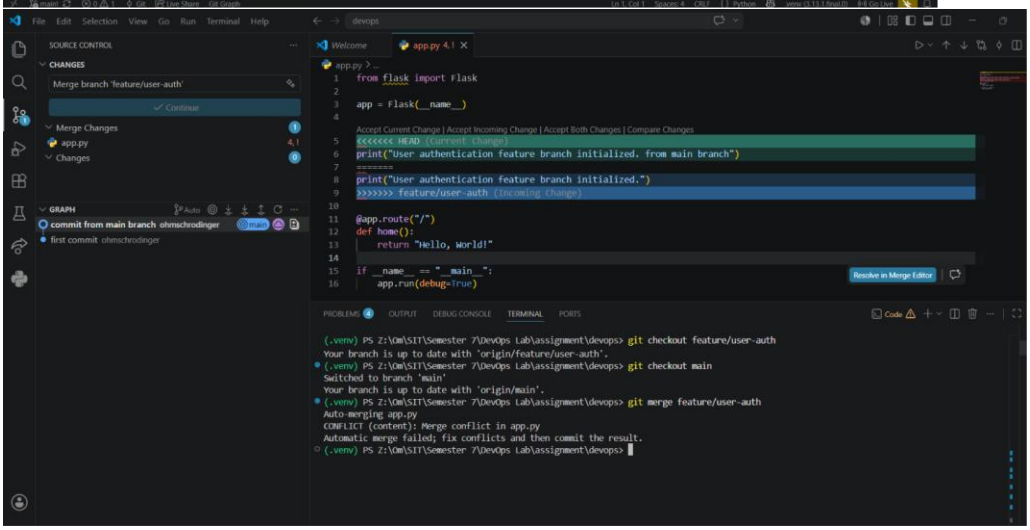


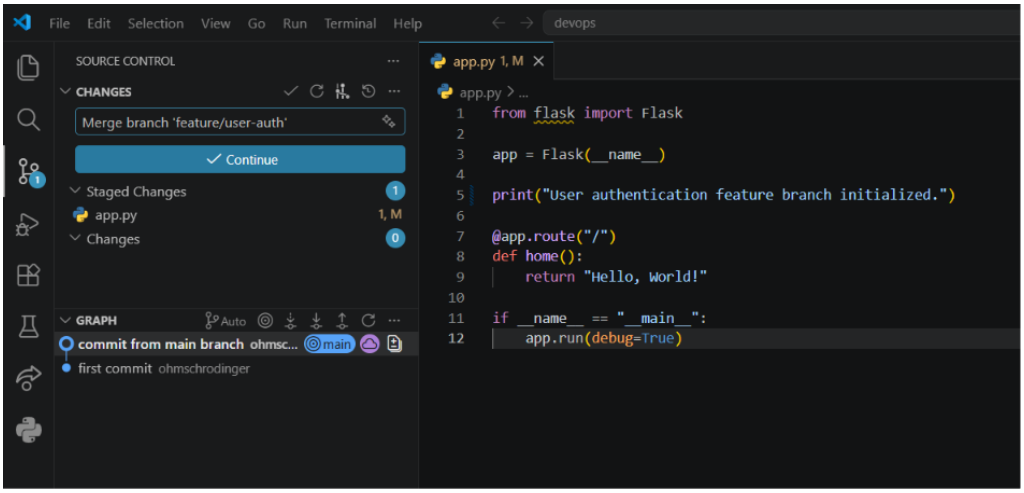
- **Task 1.3:** Simulate a conflict: On the main branch, modify the *same line* you changed in feature/user-auth. Commit this change. Now, attempt to merge feature/user-auth into main. Resolve the merge conflict manually and commit the resolution. Push the updated main branch. (1.5 Marks)

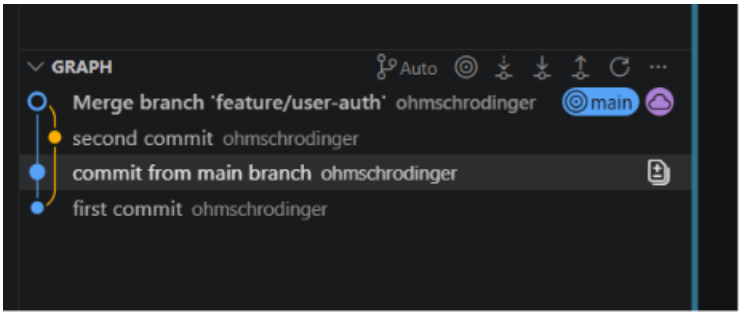



File Edit Selection View Go Run Terminal Help
SOURCE CONTROL
CHANGES
commit from main branch
Commit
Changes
app.py
1, M
GRAPH
Auto
first commit ohmschrodinger
main

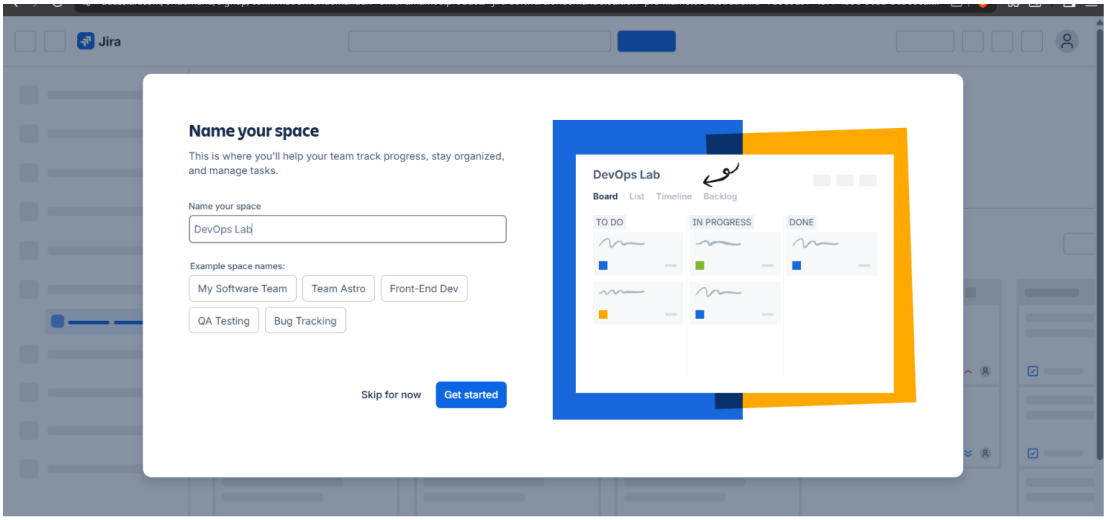

Merging: app.py 1, 1
Incoming
Current
Result app.py
from flask import Flask
app = Flask(__name__)
print("User authentication feature branch initialized.")
print("User authentication feature branch initialized. from main branch")
@app.route("/")
def home():
return "Hello, World!"
if __name__ == "__main__":
app.run(debug=True)
Complete Merge
(.venv) PS Z:\Om\SIT\Semester 7\DevOps Lab\assignment\devops> git checkout main
(.venv) PS Z:\Om\SIT\Semester 7\DevOps Lab\assignment\devops> git merge feature/user-auth
Auto-merging app.py
CONFLICT (content): Merge conflict in app.py
Automatic merge failed; fix conflicts and then commit the result.

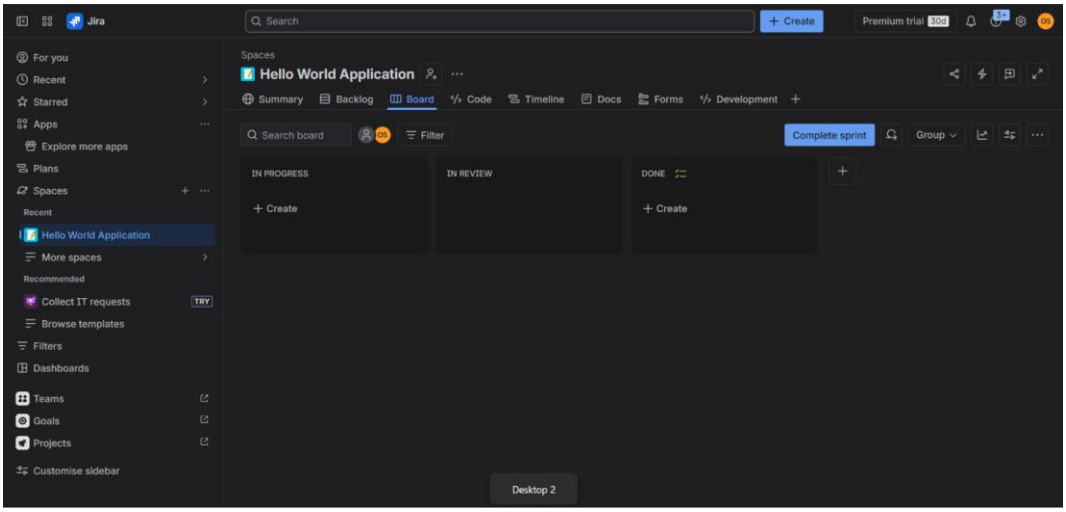

Merge branch 'feature/user-auth'
Continue
Merge Changes
app.py
Changes
GRAPH
commit from main branch ohmschrodinger
first commit ohmschrodinger
app.py 4, 1
from flask import Flask
app = Flask(__name__)
<<<<<<< HEAD (Current Change)
print("User authentication feature branch initialized. from main branch")
=======
print("User authentication feature branch initialized.")
>>>>>>> feature/user-auth (Incoming Change)
@app.route("/")
def home():
return "Hello, World!"
if __name__ == "__main__":
app.run(debug=True)
Resolve in Merge Editor
(.venv) PS Z:\Om\SIT\Semester 7\DevOps Lab\assignment\devops> git checkout feature/user-auth
Your branch is up to date with 'origin/feature/user-auth'.
(.venv) PS Z:\Om\SIT\Semester 7\DevOps Lab\assignment\devops> git checkout main
Switched to branch 'main'
Your branch is up to date with 'origin/main'.
(.venv) PS Z:\Om\SIT\Semester 7\DevOps Lab\assignment\devops> git merge feature/user-auth
Auto-merging app.py
CONFLICT (content): Merge conflict in app.py
Automatic merge failed; fix conflicts and then commit the result.
(.venv) PS Z:\Om\SIT\Semester 7\DevOps Lab\assignment\devops>



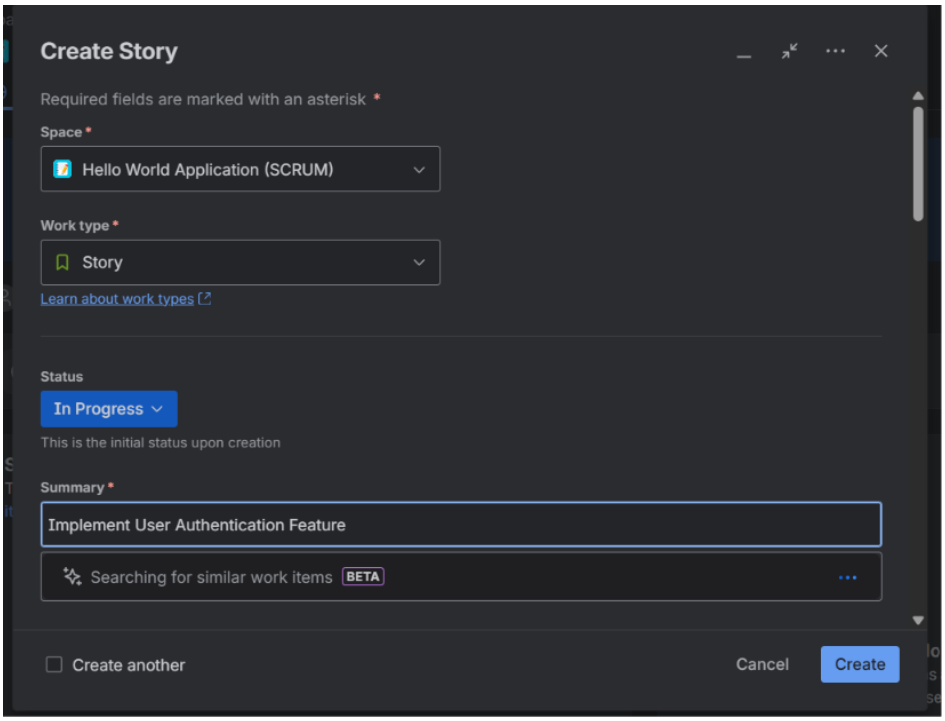


2. **Jira Project & Issue Tracking (3 Marks)**
   - **Task 2.1:** Log in to your free Jira Cloud instance. Create a new "Scrum" project for the "Hello World" application. (1 Mark)

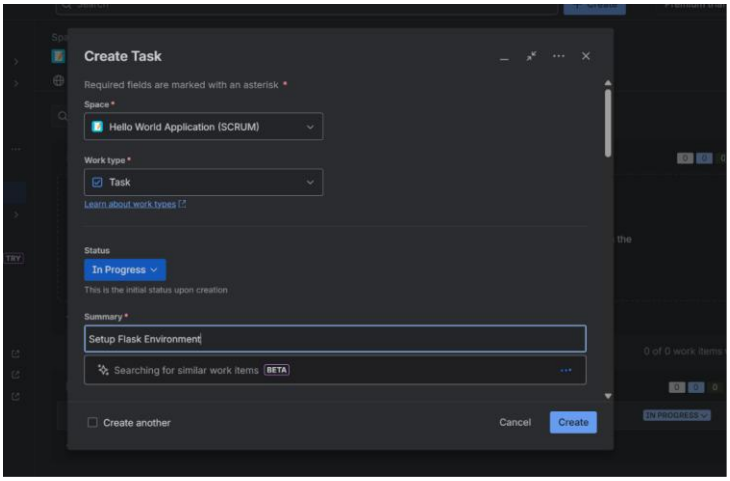




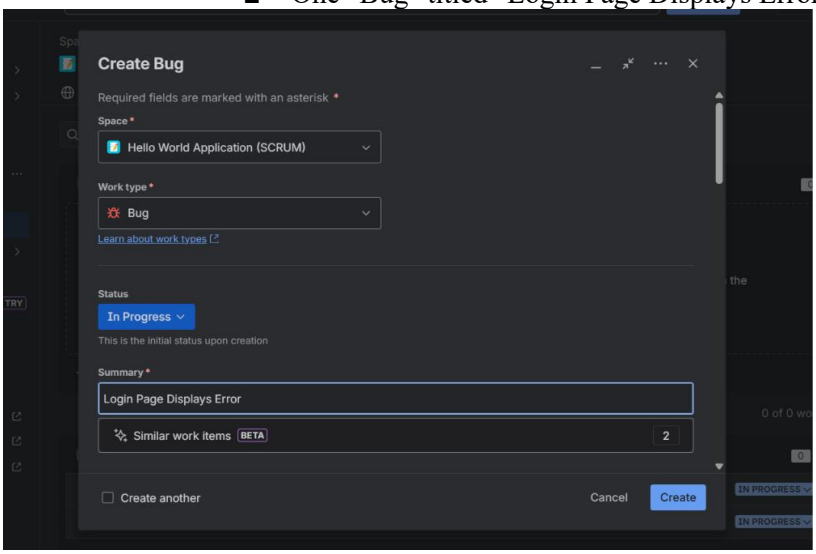
- **Task 2.2:** Create the following issues within your new project:
  - One "Story" titled "Implement User Authentication Feature".



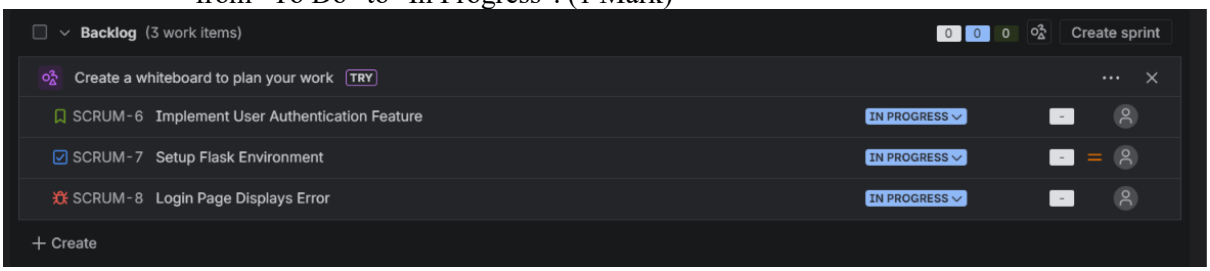
  - One "Task" titled "Setup Flask Environment".



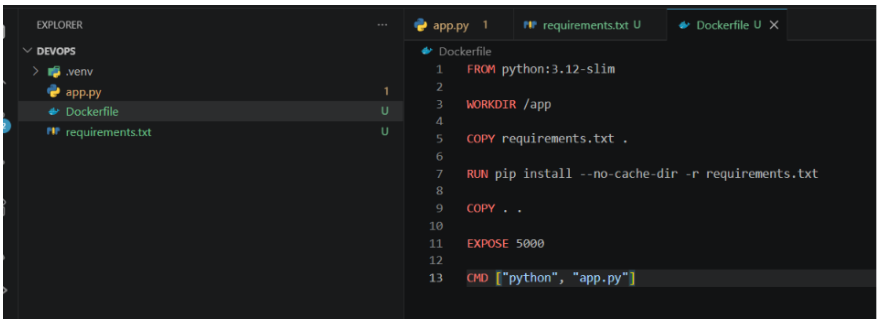
- One "Bug" titled "Login Page Displays Error". (1 Mark)

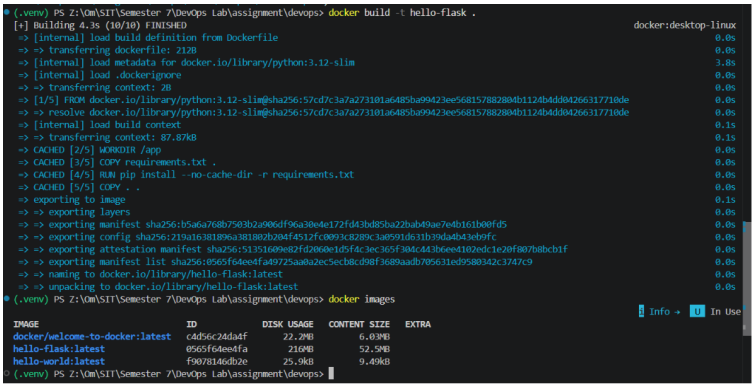


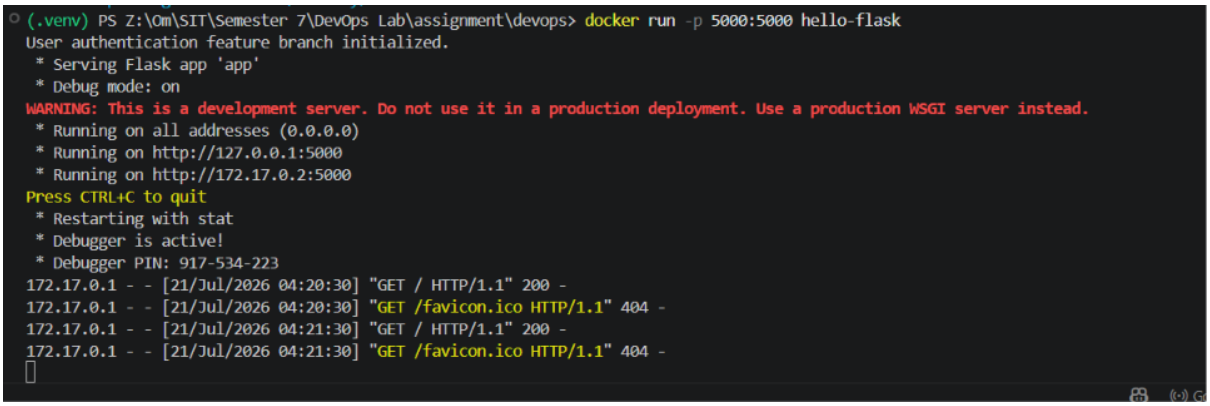
- **Task 2.3:** Navigate to the project's board. Move the "Setup Flask Environment" task from "To Do" to "In Progress". (1 Mark)



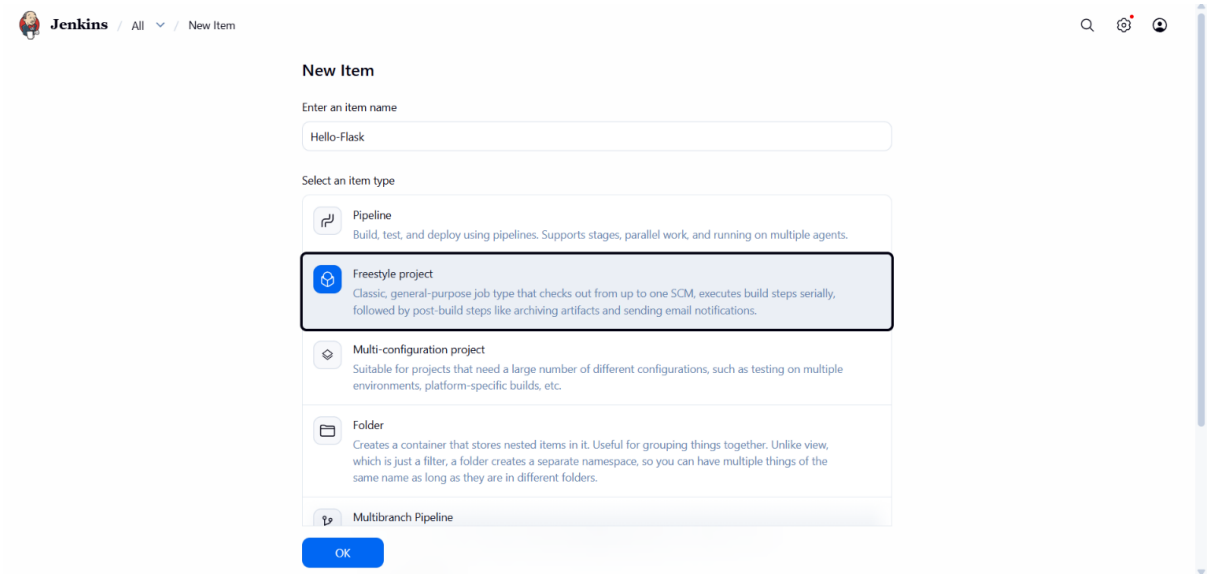
3. **Basic Containerization (Docker) & Jenkins Freestyle Project (3 Marks)**
   - **Task 3.1:** Create a Dockerfile for your Python Flask "Hello World" application. The Dockerfile should build a minimal image that runs the application on port 5000. Build the Docker image locally and verify it runs correctly. (1.5 Marks)

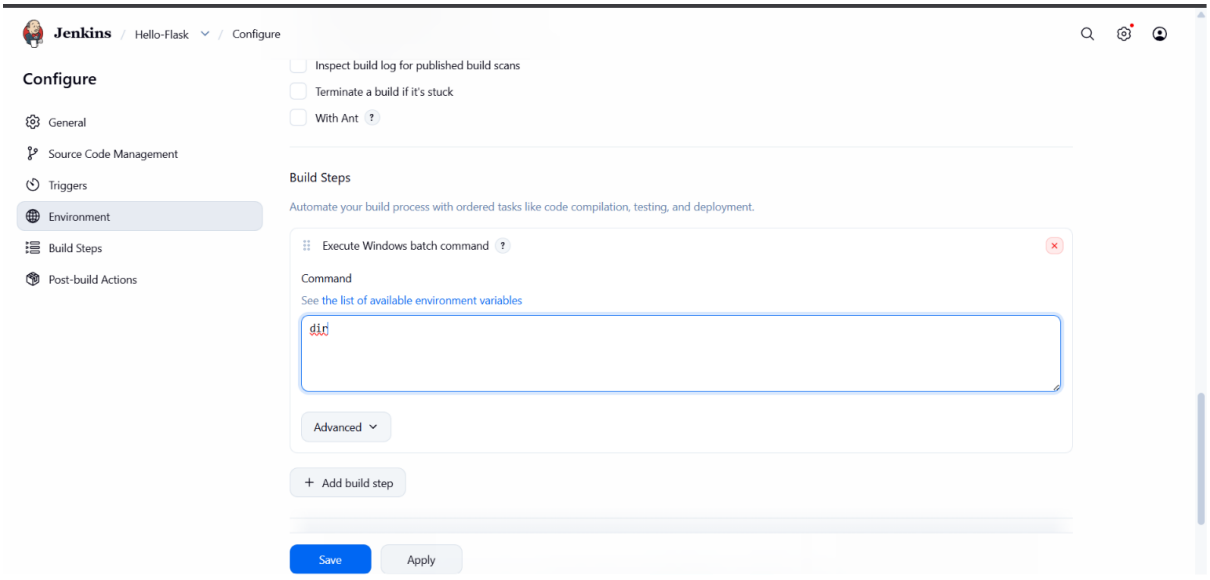




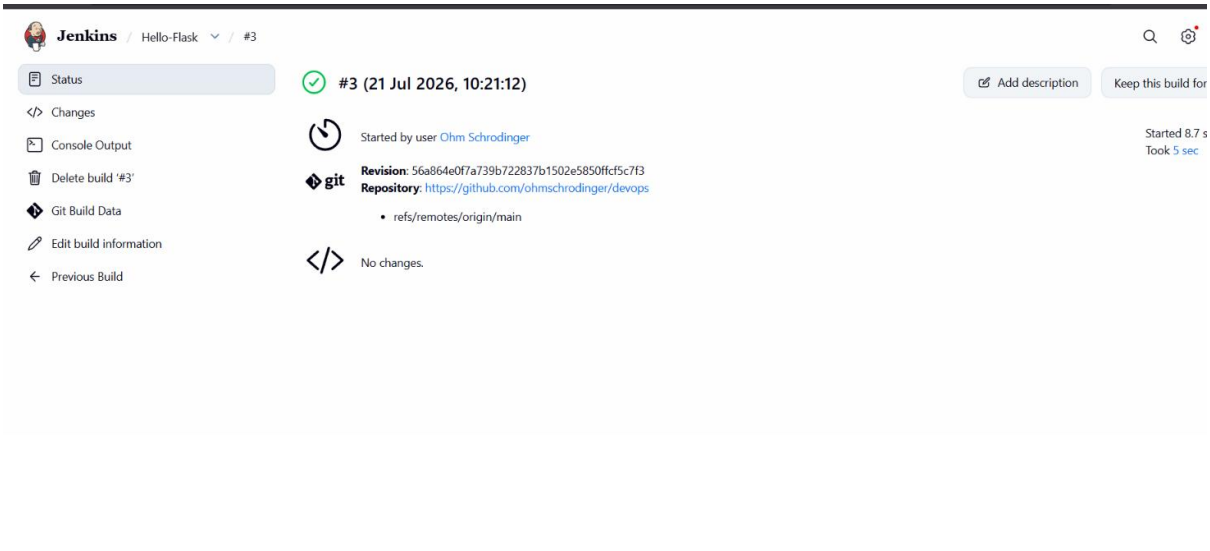


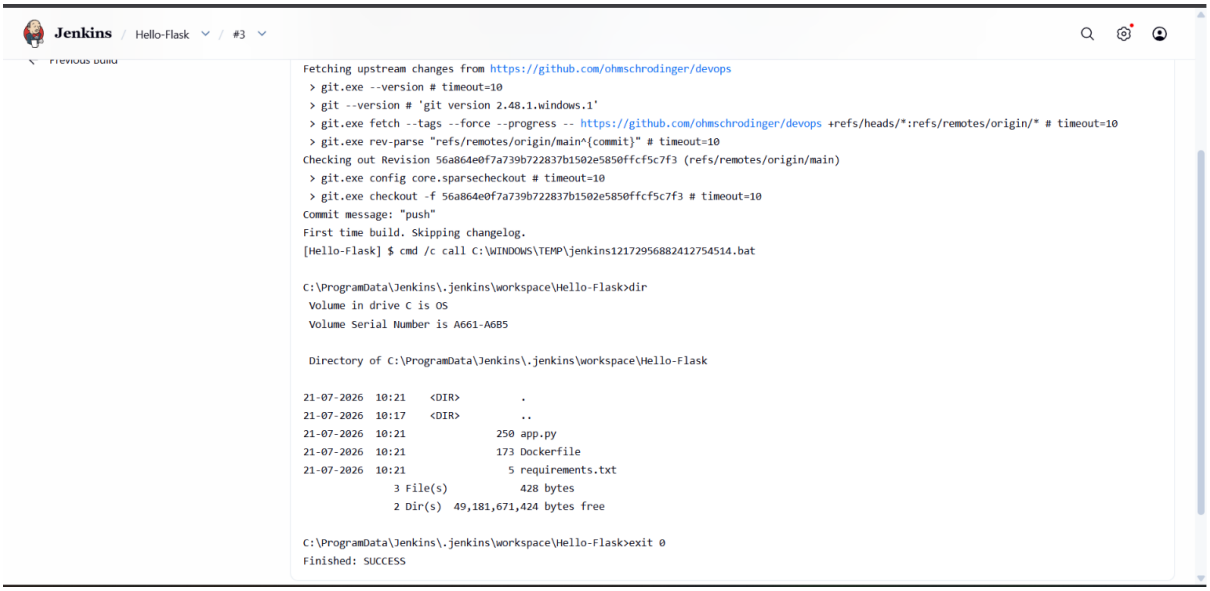
- **Task 3.2:** Set up a Jenkins Freestyle project. Configure it to pull your Git repository (from Task 1.1) and execute a build step that lists the contents of the workspace. Trigger a build and provide a screenshot of the successful build output. (1.5 Marks)


Jenkins / All / New Item
New Item
Enter an item name
Hello-Flask
Select an item type
Pipeline
Build, test, and deploy using pipelines. Supports stages, parallel work, and running on multiple agents.
Freestyle project
Classic, general-purpose job type that checks out from up to one SCM, executes build steps serially, followed by post-build steps like archiving artifacts and sending email notifications.
Multi-configuration project
Suitable for projects that need a large number of different configurations, such as testing on multiple environments, platform-specific builds, etc.
Folder
Creates a container that stores nested items in it. Useful for grouping things together. Unlike view, which is just a filter, a folder creates a separate namespace, so you can have multiple things of the same name as long as they are in different folders.
Multibranch Pipeline
OK


Jenkins / Hello-Flask / Configure
Configure
General
Source Code Management
Triggers
Environment
Build Steps
Post-build Actions
Inspect build log for published build scans
Terminate a build if it's stuck
With Ant
Build Steps
Automate your build process with ordered tasks like code compilation, testing, and deployment.
Execute Windows batch command
Command
See the list of available environment variables
dir
Advanced
Add build step
Save
Apply


Build Now
Delete Project
Configure
Rename
Builds
Filter
Today
#3 10:21
#2 10:19
#1 10:17
Last build (#3), 6 min 50 sec ago
Last stable build (#3), 6 min 50 sec ago
Last successful build (#3), 6 min 50 sec ago
Last failed build (#2), 8 min 12 sec ago
Last unsuccessful build (#2), 8 min 12 sec ago
Last completed build (#3), 6 min 50 sec ago
Build scheduled


Jenkins
Hello-Flask
#3
Status
Changes
Console Output
Delete build '#3'
Git Build Data
Edit build information
Previous Build
#3 (21 Jul 2026, 10:21:12)
Add description
Keep this build for
Started by user Ohm Schrodinger
Started 8.7 s
Took 5 sec
Revision: 56a864e0f7a739b722837b1502e5850ffcf5c7f3
Repository: https://github.com/ohmschrodinger/devops
refs/remotes/origin/main
No changes.


Jenkins
Hello-Flask
#3
Fetching upstream changes from https://github.com/ohmschrodinger/devops
> git.exe --version # timeout=10
> git --version # 'git version 2.48.1.windows.1'
> git.exe fetch --tags --force --progress -- https://github.com/ohmschrodinger/devops +refs/heads/*:refs/remotes/origin/* # timeout=10
> git.exe rev-parse "refs/remotes/origin/main^{commit}" # timeout=10
Checking out Revision 56a864e0f7a739b722837b1502e5850ffcf5c7f3 (refs/remotes/origin/main)
> git.exe config core.sparsecheckout # timeout=10
> git.exe checkout -f 56a864e0f7a739b722837b1502e5850ffcf5c7f3 # timeout=10
Commit message: "push"
First time build. Skipping changelog.
[Hello-Flask] $ cmd /c call C:\WINDOWS\TEMP\jenkins12172956882412754514.bat
C:\ProgramData\Jenkins\.jenkins\workspace\Hello-Flask>dir
Volume in drive C is OS
Volume Serial Number is A661-A6B5
Directory of C:\ProgramData\Jenkins\.jenkins\workspace\Hello-Flask
21-07-2026 10:21 <DIR> .
21-07-2026 10:17 <DIR> ..
21-07-2026 10:21 250 app.py
21-07-2026 10:21 173 Dockerfile
21-07-2026 10:21 5 requirements.txt
3 File(s) 428 bytes
2 Dir(s) 49,181,671,424 bytes free
C:\ProgramData\Jenkins\.jenkins\workspace\Hello-Flask>exit 0
Finished: SUCCESS

2. TW2 - Continuous Evaluation: Advanced DevOps Practices (10 Marks)

Instructions: Build upon your previous work. Document your commands, configuration files (e.g., docker-compose.yml, Kubernetes YAMLs), and outputs.

Scenario: You need to enhance your "Hello World" application to include a database and deploy it using orchestration tools.

1. **Docker Compose for Multi-Container App (3 Marks)**
    - **Task 1.1:** Modify your "Hello World" Flask application to include a simple PostgreSQL database (e.g., the app connects to and stores a single entry). (1 Mark)

app.py 2, M | requirements.txt M | Dockerfile | docker-compose.yml U

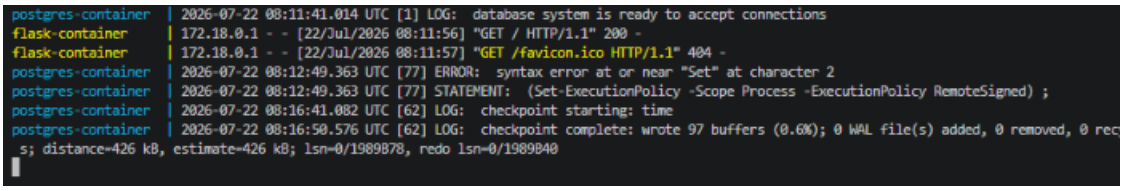
app.py > home

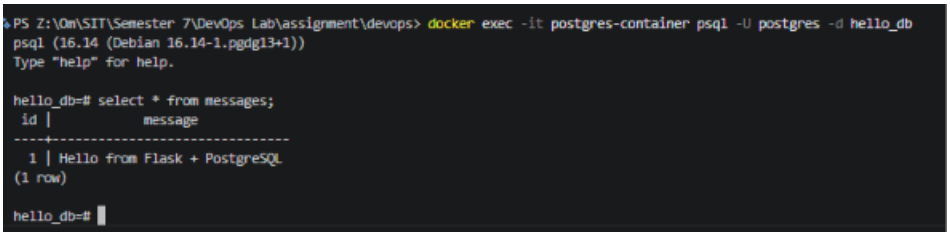
```python
from flask import Flask
import psycopg2
import time

app = Flask(__name__)

print("User authentication feature branch initialized.")


def get_db_connection():

    while True:
        try:
            conn = psycopg2.connect(
                host="db",
                database="hello_db",
                user="postgres",
                password="password"
            )
            return conn

        except Exception:
            print("Database not ready, waiting...")
            time.sleep(2)


@app.route("/")
def home():

    conn = get_db_connection()

    cursor = conn.cursor()

    cursor.execute("""
        CREATE TABLE IF NOT EXISTS messages(
            id SERIAL PRIMARY KEY,
            message TEXT
        )
    """)


    cursor.execute(
        "INSERT INTO messages(message) VALUES(%s)",
        ("Hello from Flask + PostgreSQL",)
    )


    conn.commit()

    cursor.close()
    conn.close()

    return "Hello World! Data stored in PostgreSQL."
```

```python
if __name__ == "__main__":

    app.run(
        host="0.0.0.0",
        port=5000,
        debug=True
    )
```

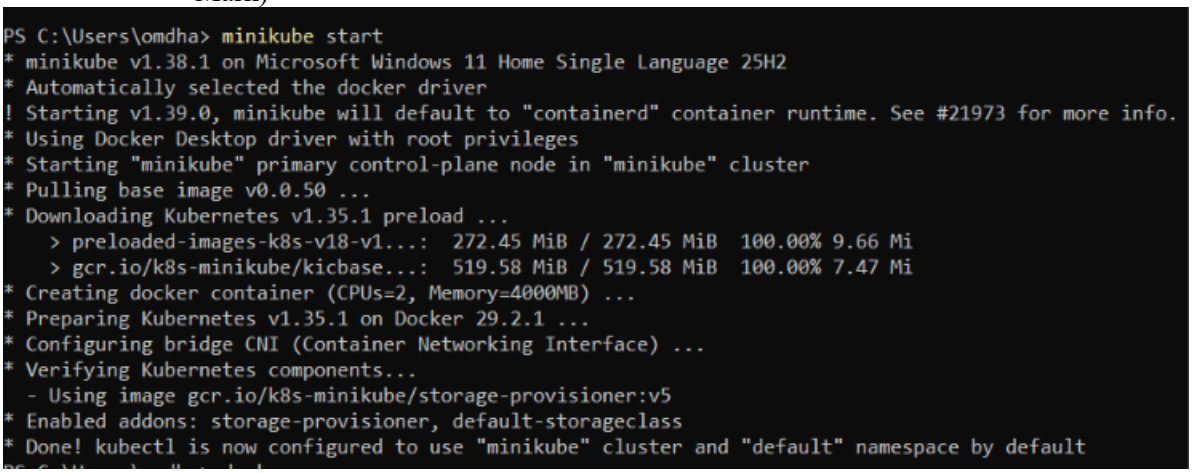
- **Task 1.2:** Create a docker-compose.yml file that defines two services: your Flask application and the PostgreSQL database. Configure them to communicate. Bring up the

application using docker-compose up and verify both services are running and communicating. Provide the docker-compose.yml and output. (2 Marks)





2. **Kubernetes Deployment & Service (4 Marks)**
   - **Task 2.1:** Ensure your Docker image from TW1 (or an updated one) is accessible (e.g., pushed to a public Docker Hub repository, or Minikube's Docker daemon is used). (1 Mark)



```
PS C:\Users\omdha> docker ps
CONTAINER ID   IMAGE                                 COMMAND                  CREATED          STATUS          PORTS
                              NAMES
ebeab3eda438   gcr.io/k8s-minikube/kicbase:v0.0.50   "/usr/local/bin/entr…"   4 minutes ago    Up 4 minutes    127.0.0.1:49965->22/tcp, 127.0.0.1:49963->2376/tcp, 127.0.0.1:49964->5000/tcp, 127.0.0.1:49967->8443
/tcp, 127.0.0.1:49966->32443/tcp   minikube
11fa414e9dca   devops-flask-app                      "python app.py"          27 minutes ago   Up 27 minutes   0.0.0.0:5000->5000/tcp, [::]:5000->5000/tcp
                              flask-container
6ec9bf009dd2   postgres:16                           "docker-entrypoint.s…"   27 minutes ago   Up 27 minutes   0.0.0.0:5432->5432/tcp, [::]:5432->5432/tcp
                              postgres-container
```

```
PS C:\Users\omdha> kubectl cluster-info
Kubernetes control plane is running at https://127.0.0.1:49967
CoreDNS is running at https://127.0.0.1:49967/api/v1/namespaces/kube-system/services/kube-dns:dns/proxy

To further debug and diagnose cluster problems, use 'kubectl cluster-info dump'.
PS C:\Users\omdha> kubectl get nodes
NAME       STATUS   ROLES           AGE    VERSION
minikube   Ready    control-plane   5m5s   v1.35.1
PS C:\Users\omdha> docker images

IMAGE                                                                                                     ID             DISK USAGE   CONTENT SIZE   EXTRA
devops-flask-app:latest                                                                                   3bfbaf1d905e        232MB           57MB    U
docker/welcome-to-docker:latest                                                                           c4d56c24da4f       22.2MB         6.03MB
gcr.io/k8s-minikube/kicbase:v0.0.50                                                                       b97074569ae9       1.94GB          545MB
gcr.io/k8s-minikube/kicbase@sha256:eb4fec00e8ad70adf8e6436f195cc429825ffb85f95afcdb5d8d9deb576f3e93       eb4fec00e8ad       1.94GB          545MB    U
hello-flask:latest                                                                                        a45ba7fa2d28        216MB         52.5MB    U
hello-world:latest                                                                                        f9078146db2e       25.9kB         9.49kB
postgres:16                                                                                               33f923b05f64        642MB          166MB    U
PS C:\Users\omdha> minikube image load devops-flask-app:latest
PS C:\Users\omdha>
```

- **Task 2.2:** Create a Kubernetes Deployment YAML for your Flask application. Deploy it to your local Kubernetes cluster (Minikube/Kind). Verify the Pods are running. (1.5 Marks)

deployment.yaml > {} spec > {} template > {} spec > {} containers > {} flask-container > {} ports > {} [0] > containerPort

```yaml
apiVersion: apps/v1
kind: Deployment

metadata:
  name: flask-deployment

spec:
  replicas: 2

  selector:
    matchLabels:
      app: flask

  template:

    metadata:
      labels:
        app: flask

    spec:

      containers:

      - name: flask-container

        image: devops-flask-app:latest

        imagePullPolicy: Never

        ports:
        - containerPort: 5000
```

```
PS Z:\Om\SIT\Semester 7\DevOps Lab\assignment\devops> kubectl apply -f deployment.yaml
deployment.apps/flask-deployment created
PS Z:\Om\SIT\Semester 7\DevOps Lab\assignment\devops> kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
flask-deployment-84964d4d6b-2p662   1/1     Running   0          55s
flask-deployment-84964d4d6b-lbncm   1/1     Running   0          55s
PS Z:\Om\SIT\Semester 7\DevOps Lab\assignment\devops>
```

- **Task 2.3:** Create a Kubernetes Service YAML (type NodePort or LoadBalancer if supported) to expose your Flask application. Apply the manifest and provide the command to access your application from your host machine. (1.5 Marks)

service.yaml > { } spec > { } ports > { } [0] > nodePort

```yaml
apiVersion: v1
kind: Service

metadata:
  name: flask-service

spec:

  type: NodePort

  selector:
    app: flask

  ports:

  - port: 5000
    targetPort: 5000
    nodePort: 30007
```

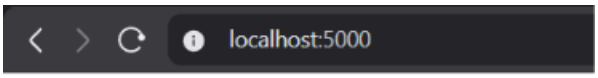
```
PS Z:\Om\SIT\Semester 7\DevOps Lab\assignment\devops> kubectl apply -f service.yaml
service/flask-service created
PS Z:\Om\SIT\Semester 7\DevOps Lab\assignment\devops> kubectl get services
NAME            TYPE        CLUSTER-IP      EXTERNAL-IP   PORT(S)          AGE
flask-service   NodePort    10.109.217.59   <none>        5000:30007/TCP   9s
kubernetes      ClusterIP   10.96.0.1       <none>        443/TCP          11m
PS Z:\Om\SIT\Semester 7\DevOps Lab\assignment\devops> minikube service flask-service
```

NAMESPACE	NAME	TARGET PORT	URL
default	flask-service	5000	http://192.168.49.2:30007

```
* Starting tunnel for service flask-service.
```

NAMESPACE	NAME	TARGET PORT	URL
default	flask-service		http://127.0.0.1:60022

```
* Opening service default/flask-service in default browser...
! Because you are using a Docker driver on windows, the terminal needs to be open to run it.
```



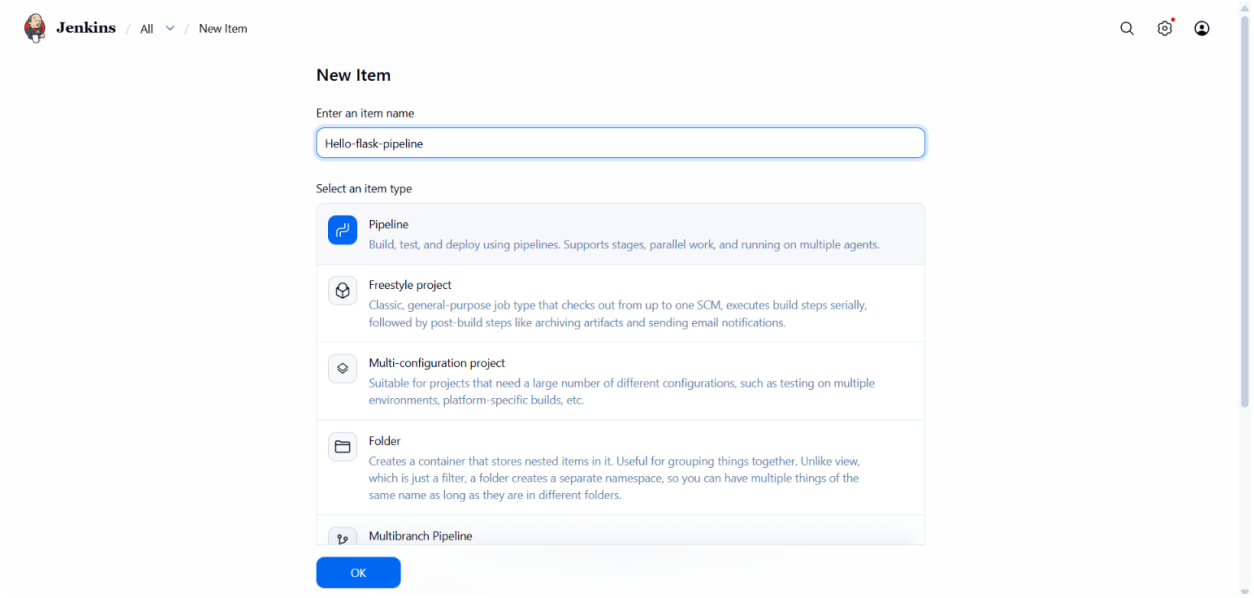
Hello World! Data stored in PostgreSQL.

3. **Integrated CI/CD with Jenkins & Basic Infrastructure as Code (IaC) (3 Marks)**
   - **Task 3.1:** Create a Jenkins Declarative Pipeline that performs the following steps:
     - Checks out your Git repository.

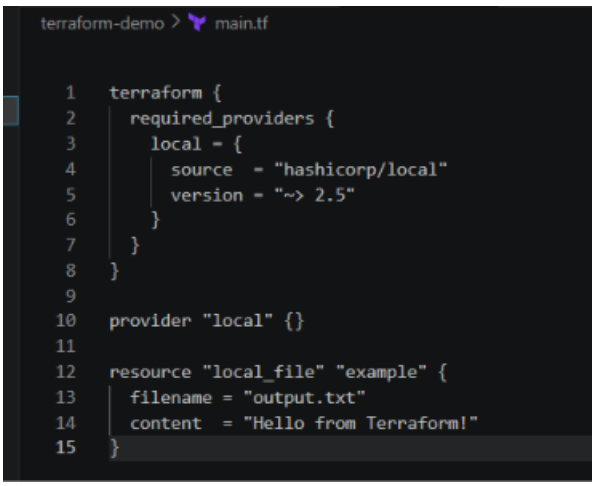
- Builds the Docker image for your Flask application (from Task 1.1).
- (Optional but Recommended for higher scores): Pushes the built Docker image to a registry. (2 Marks)

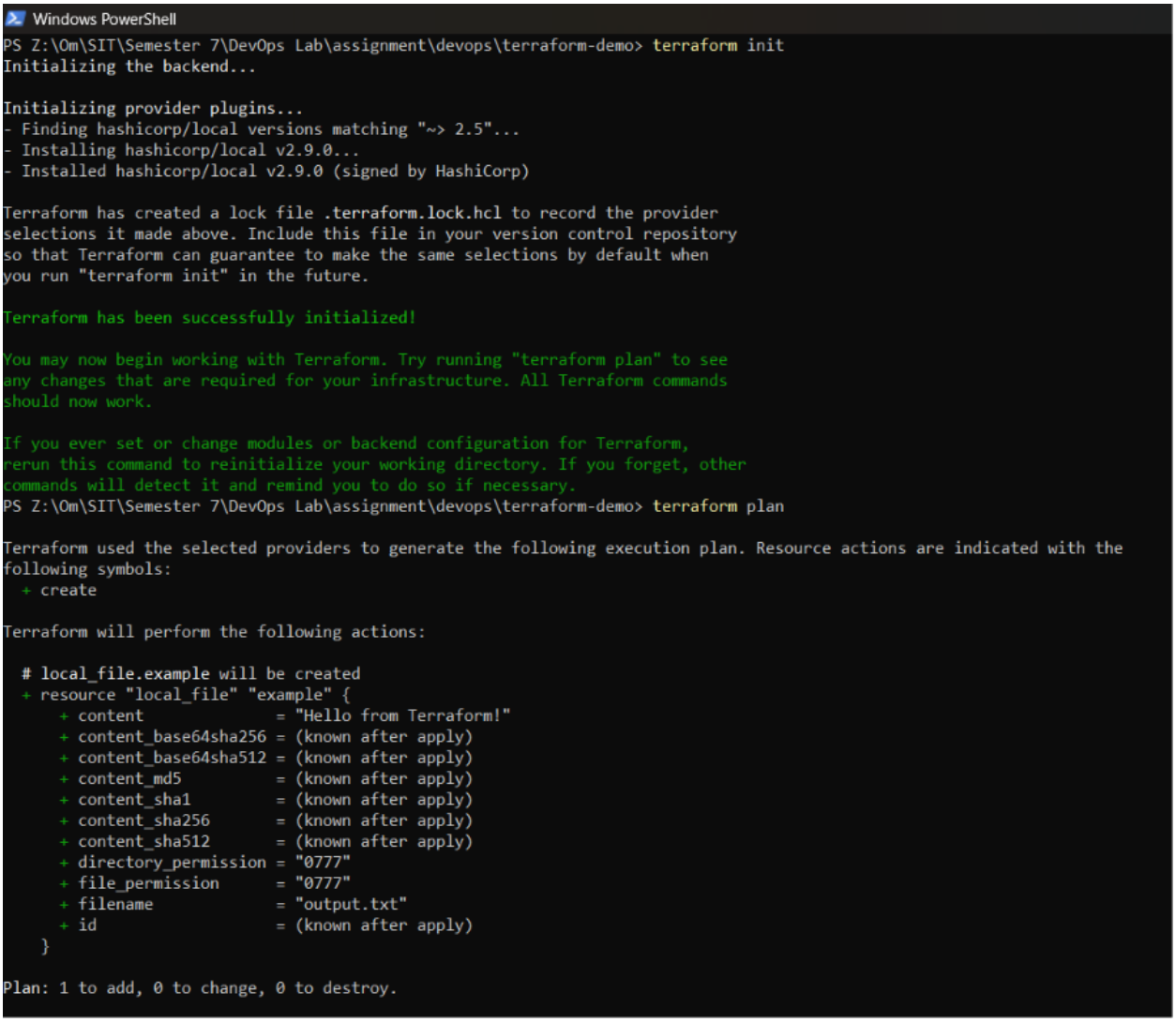
Jenkinsfile

```
pipeline {
    agent any

    stages {

        stage('Checkout') {
            steps {
                git 'https://github.com/ohmschrodinger/devops.git'
            }
        }

        stage('Build Docker Image') {
            steps {
                bat 'docker build -t devops-flask-app .'
            }
        }

        stage('List Docker Images') {
            steps {
                bat 'docker images'
            }
        }

        // Optional
        stage('Push Docker Image') {
            when {
                expression { false }   // Change to true after docker login
            }
            steps {
                bat 'docker push ohmschrodinger/devops-flask-app:latest'
            }
        }
    }
}
```



- **Task 3.2:** Install Terraform. Write a simple Terraform configuration (main.tf) to create a local file (e.g., output.txt) with some content. Initialize, plan, and apply the configuration. Show the output of terraform plan and verify the file creation. (1 Mark)




Windows PowerShell
PS Z:\Om\SIT\Semester 7\DevOps Lab\assignment\devops\terraform-demo> terraform init
Initializing the backend...
Initializing provider plugins...
- Finding hashicorp/local versions matching "~> 2.5"...
- Installing hashicorp/local v2.9.0...
- Installed hashicorp/local v2.9.0 (signed by HashiCorp)
Terraform has created a lock file .terraform.lock.hcl to record the provider
selections it made above. Include this file in your version control repository
so that Terraform can guarantee to make the same selections by default when
you run "terraform init" in the future.
Terraform has been successfully initialized!
You may now begin working with Terraform. Try running "terraform plan" to see
any changes that are required for your infrastructure. All Terraform commands
should now work.
If you ever set or change modules or backend configuration for Terraform,
rerun this command to reinitialize your working directory. If you forget, other
commands will detect it and remind you to do so if necessary.
PS Z:\Om\SIT\Semester 7\DevOps Lab\assignment\devops\terraform-demo> terraform plan
Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the
following symbols:
+ create
Terraform will perform the following actions:
# local_file.example will be created
+ resource "local_file" "example" {
+ content = "Hello from Terraform!"
+ content_base64sha256 = (known after apply)
+ content_base64sha512 = (known after apply)
+ content_md5 = (known after apply)
+ content_sha1 = (known after apply)
+ content_sha256 = (known after apply)
+ content_sha512 = (known after apply)
+ directory_permission = "0777"
+ file_permission = "0777"
+ filename = "output.txt"
+ id = (known after apply)
}
Plan: 1 to add, 0 to change, 0 to destroy.

```
Note: You didn't use the -out option to save this plan, so Terraform can't guarantee to take exactly these actions if
you run "terraform apply" now.
PS Z:\Om\SIT\Semester 7\DevOps Lab\assignment\devops\terraform-demo> terraform apply

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the
following symbols:
  + create

Terraform will perform the following actions:

  # local_file.example will be created
  + resource "local_file" "example" {
      + content              = "Hello from Terraform!"
      + content_base64sha256 = (known after apply)
      + content_base64sha512 = (known after apply)
      + content_md5          = (known after apply)
      + content_sha1         = (known after apply)
      + content_sha256       = (known after apply)
      + content_sha512       = (known after apply)
      + directory_permission = "0777"
      + file_permission      = "0777"
      + filename             = "output.txt"
      + id                   = (known after apply)
    }

Plan: 1 to add, 0 to change, 0 to destroy.

Do you want to perform these actions?
  Terraform will perform the actions described above.
  Only 'yes' will be accepted to approve.

  Enter a value: yes

local_file.example: Creating...
local_file.example: Creation complete after 0s [id=cadbafd9572a72dc7769c97730552f5955e135fd]

Apply complete! Resources: 1 added, 0 changed, 0 destroyed.
PS Z:\Om\SIT\Semester 7\DevOps Lab\assignment\devops\terraform-demo> type output.txt
Hello from Terraform!
PS Z:\Om\SIT\Semester 7\DevOps Lab\assignment\devops\terraform-demo>
```

3. TW3 - ESE Lab Exam + Project Evaluation + Viva (30 Marks)

Instructions: This is a comprehensive assessment that combines practical lab work with a project implementation and a viva-voce.

A. ESE Lab Exam (10 Marks)

Scenario: You are tasked with setting up a fully functional CI/CD pipeline for a given multi-service application (e.g., a simple microservice architecture with a frontend, backend, and database).

- **Task:** Given a multi-service application (frontend, backend, database components), implement a complete CI/CD pipeline from scratch.
    - Initialize a Git repository and set up a basic branching strategy.
    - Utilize Jira to create and track issues related to the deployment.
    - Develop a Jenkins pipeline that:
        - Pulls code from Git.
        - Builds Docker images for each service.
        - Pushes images to a Docker registry.
        - Deploys the services to a local Kubernetes cluster.

    - Expose the application services via Kubernetes Services.
    - (Optional for higher scores): Use a simple Terraform configuration to provision a basic infrastructure component (e.g., a local directory for logging).

Evaluation Rubric (Lab Exam - 10 Marks):

- **Problem Solving & Practical Application (5 Marks):** Ability to correctly set up, configure, and integrate all required tools to achieve the functional pipeline.
- **Accuracy & Efficiency of Solution (3 Marks):** The pipeline should run smoothly without manual intervention, and the deployed application should be fully functional.
- **Troubleshooting & Debugging (2 Marks):** Ability to quickly identify and resolve any issues encountered during the setup and execution.

B. Project Evaluation (10 Marks)

Instructions: Develop a small-scale DevOps project that demonstrates your understanding and integration of at least 4-5 core DevOps tools (e.g., Git, Jira, Jenkins, Docker, Kubernetes, Terraform, Prometheus/Grafana for monitoring, Ansible for configuration management).

Project Idea (Example): Build a "Personal Blog Application" with a CI/CD pipeline.

- **Requirements:**
    - The application should be version-controlled using Git.
    - Use Jira for project and issue tracking.
    - Implement a Jenkins pipeline for continuous integration and continuous delivery.
    - Containerize the application using Docker.
    - Deploy the containerized application to a Kubernetes cluster.
    - Use Terraform for provisioning at least one infrastructure component (e.g., a persistent volume for the blog data).
    - (Optional for higher scores): Implement basic monitoring using Prometheus/Grafana or automated testing.

Evaluation Rubric (Project - 10 Marks):

- **Project Design & Architecture (3 Marks):** How well are the DevOps tools integrated? Is the architecture sound and scalable?
- **Functionality & Completeness (3 Marks):** Does the project work as intended? Are all specified requirements met?
- **Code Quality & Best Practices (2 Marks):** Readability, comments, adherence to best practices for Dockerfiles, Jenkinsfiles, Kubernetes YAMLs, and Terraform configurations.
- **Documentation & Presentation (2 Marks):** Clarity and completeness of project documentation (e.g., README.md, setup instructions, architecture diagrams if any) and effectiveness of the project demonstration.

C. Viva (10 Marks)

Instructions: Be prepared to answer questions related to your ESE Lab Exam and Project, as well as general DevOps concepts.

Evaluation Rubric (Viva - 10 Marks):

- **Understanding of Concepts (4 Marks):** Ability to clearly explain the theoretical concepts behind the tools used, the CI/CD pipeline, containerization, orchestration, and IaC.
- **Problem-Solving & Critical Thinking (3 Marks):** Ability to discuss challenges faced during the lab/project and how they were overcome, as well as to propose alternative solutions.
- **Clarity of Explanation & Communication (3 Marks):** Articulation and confidence in explaining technical details and answering questions.